

SUMANTH GOWDA V

ELECTRONICS AND COMMUNICATION ENGINEERING

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Career Objective:

To work in challenging and responsible position in a dynamic programming organization through dedicated service with sincerely and hardworking. To enhance my knowledge with the latest technologies.

Summary:

- A budding professional working as a Hardware Test Engineer deputed to ISRO SATELLITE CENTRE from ANANTH TECHNOLOGIES LTD.
- Have 2.8 years experience in **hardware testing**.
- Possess strong knowledge and understanding of hardware Testing.
- An effective communicator with strong analytical / logical skills and ability to relate to people at any level of business and management.
- A quick learner with the ability to work under pressure and meet deadlines.
- Team player with strong analytical and leadership skills.

Education Qualification:

Course	Board/University	Institution	Year of passing	Marks
B.E ECE	Visvesvaraya Technological University (VTU)	Sri Venkateshwara College of Engineering Bangalore	2022	7.4 (CGPA)
II P.U.C (PCMB)	Department of Pre University Education Karnataka	VIDHYADRI PU COLLEGE	2018	83.8%
S.S.L.C	Karnataka Secondary Education Examination Board	SUNRISE INTERNATIONAL SCHOOL	2016	86.8%

Technical skills with Roles and Responsibilities:

- Good knowledge in testing **POWER Packages** (CORE POWER, SHUNT, DISTRIBUTION, MEPU, CURRENT SENSORS & FUSE MODULES)
- Good knowledge in analyzing the multilayer layout of PCB and blue prints.
- Problem solving at different phases of test.
- Testing experience on the **Automated Test Equipment (ATE)** based systems.
- Have knowledge on different instruments handled by the avionics division.
- Knowledge on different electronic components involved in the flight model PCB cards.
- Functionality testing of PCB when all the electronic components were mounted.
- Study of test requirements and visual inspection for the functionality of the cards.
- Realization of test setup and Testing of Power cards in card level and package level in three different phases.
- Worked on test equipment's such as Digital Storage Oscilloscope, Signal generator, Spectrum Analyzer, Triple output power supply, Counter, Power meter, Audio Analyzer, Console, Function generator, Power supplies, LCR meter, Thermal imager, AVO meter, TM simulator, Electronic Load, Rheostat, Linear power supply, DC-DC Test Jig, Digital multi meter and Current probe amplifier.
- Flight card Debug the test failures, and support to ISRO engineers in debug design issues
- Experienced in System level testing & **POWER Packages** of satellite at different phases like Environmental Test which includes Vibration test & Thermovac as per MIL standards.
- Have good experience in testing harness and connections.
- Basics of MICRO CONTROLLER.
- Operating system used : Versions of windows.
- Tools Used : Labview, Matlab.

Academic projects:

1. "Smart Car Parking System"

- Technology Used : C, IOT
- Description : Internet of things (IOT) is the emerging paradigm, which contains huge amount of smart object and smart devices connected to the internet for communicating with each other.

This system collects the car parking data and it use wisely and allot the parking space to all the user comfortably. Using this system we can manage the parking by getting to know how many cars are in the parking area and how many cars are leaving parking slot. Using this information we can able to map to the user interface.

2. "Plant Growth Monitoring System"

- Technology Used : Python, Machine Learning
- Description : Plant growth monitoring system using Artificial Intelligence is project which link the technology and the farmer. Here we have developed the model which capture the image and stored in the database. Later when we train those images using a machine learning parameter, based on the the parameter they give the output weather the plant is healthy or unhealthy.

Languages Known:

Speak : English,Hindi,Kannada,Telugu.
Read : English,Hindi,Kannada.
Write : English,,Kannada.

Work Experience Details:

- Working as a Test engineer in Ananth Technologies Ltd, Bangalore from 01th SEP 2022 to till date.

#1 Project Details:

Project Name : GAGANYAAN FOR ASTRONAUTS.
Package : FOOD HEATER
Client : DRDO-DFRL (Defence Food Research Laboratory)
Role : Test Engineer

Description:

- DRDO, DFRL-Defence Food Research Laboratory has designed and developed and electrically operated Food Heater suitable for heating RTE RPP food products to consumption temperature for astronauts suiting to microgravity conditions in the space module.
- The Food heater being designed for the Gaganyaan mission will be mounted inside the space module for warming the RTE(Ready to Eat) food packets prior to consumption by the astronauts. It will utilize the power from onboard DC power supply unit having a voltage range from 28 to 42 VDC. However the RTE(Ready to eat) RPP(Retort Pouch Processed) food products need to be warmed to 60-70 degree Celsius before consumption. The system can warm the packets to their consumption temperature in reasonable time.

#2 Project Details:

Project Name : INSAT-3DS, RISAT-2A/1B, GSAT-7R,OCEANSAT-3A,GAGANYAAN, RESOURCESAT.
Package : Current Sensors,FDM , EED (Electro Explosive Device), Core power, Distribution, Shunt packages
Client : Indian Space Research Organization (ISRO)
Role : Test Engineer

Description:

- Current sensors have different PCB cards namely BCS, LCS, SAGS & MCS. All this card has different current range between 0 to 130A.These are used to measure battery charge, discharge, solar array generation & load currents. Telemetry voltage for all these ranges from -5V to +5V. The few environmental tests will do for make sure that the power card works at all temperature in space.
- EED scheme is configured to meet the safety requirements. There are 3 electrochemical breaks which are provided between Battery and Squibs.
 - EED Enable or Disable relay is common for all 3 groups.
 - Arm and Disarm relay separate for each group.
 - Event firing relay for individual squibs.

A spacecraft uses several deployment mechanisms & pyro operated propulsion elements. These electro explosive devices events require battery to power the squibbs for about 64ms duration.

- Core power unit packages have motherboard packages configuration. The core power unit is designed to provide mainly TM/TC interface between power system & TM/TC subsystem.
- Distribution packages needed for battery charging, discharging, bus formatting & distribution, Main function of power distribution package is to distribute power to various systems. Loads mainly 3 types of buses
 1. Normal power bus
 2. Un-interrupted bus
 3. Battery bus
- Shunt packages: The solar array output voltage is regulated to 42V/70V during sunlit period by sequential switching shunt regulator.

Regulating of bus voltage is achieved by sequential switching the MOSFET switches connected across each solar array strings. The bus voltage sensed is processed by a majority voltage logic to include or exclude string. 4 strings out of 16 strings are charger strings and are used to charge the battery.

Personal Profile:

Name	:	Sumanth Gowda V
Fathers Name	:	Venkatesh K
Date of birth	:	03-04-2000
Nationality	:	Indian
Religion	:	Hindu
Gender	:	Male
Marital Status	:	Single

Declaration:

I hereby declare that all the information furnished above is true to the best of my knowledge and belief.

Place: Bengaluru
Date:

Yours faithfully,
(Sumanth Gowda V)